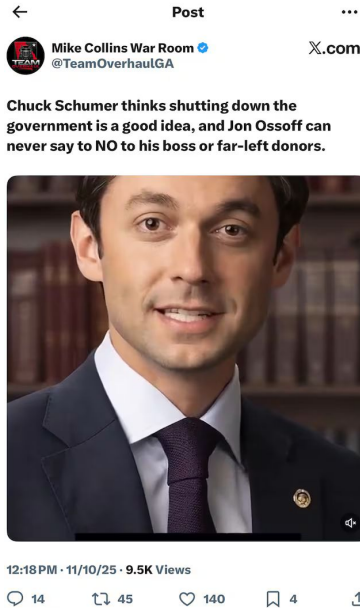




Leveraging TDA for Image Deepfake Detection

By: Srikar Kunapuli

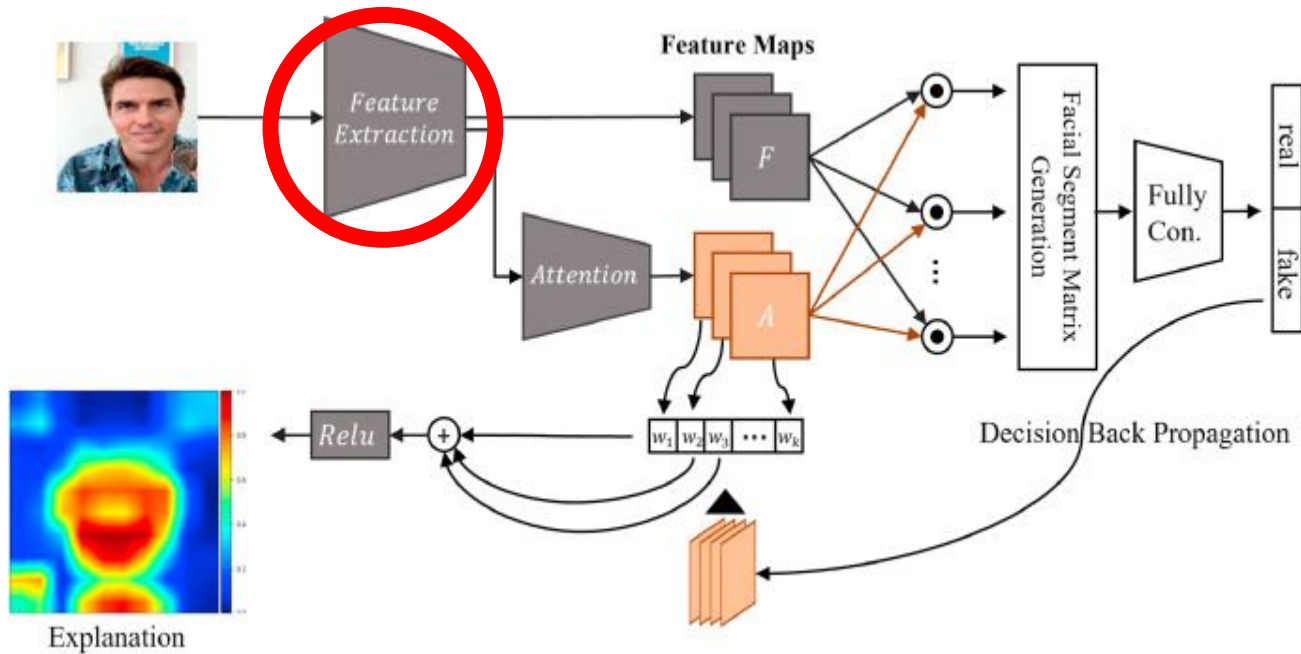
Motivation



- Loss of credibility and public trust
- Fabrication of social media
- Increasing mass polarization
- Defamatory and harmful to the victim

Source: *The Atlanta Journal-Constitution*, Patricia Murphy, Nov. 12, 2025.

Limitations of Current Models



What Features Normal Models Extract ?

- Surface pixel textures
- Face blending mistakes
- Checkerboard noise patterns
- Irregular edge boundaries



Feature Extraction via TDA

- Analyze the coarse, global structure of the image
- Detects specific image features via filtrations instead of pixel patterns
- Has higher stability through compression and image compression
- Produces interpretable features and avoids pixel overfitting



Image Structure

- Face Image Classification Dataset by Paudel (Hugging Face), containing 16,060 AI-generated and 16,060 real images
- Each grayscale image can be thought of as a 256×256 pixel space
- Each pixel is given an intensity value
- A natural complex choice is the cubical complex
- This choice allows for pixels to be “filled in” via filtration



Filtrations

- Created four filtration functions
 - **Height, Radial, Density, Dilation**
- Each filtration ranks pixels
- Each pixel enters the filtration through a filtration threshold value
- Filtrations help to transform pixel-level features into structural-image features

Filtration Results: 00000026_29

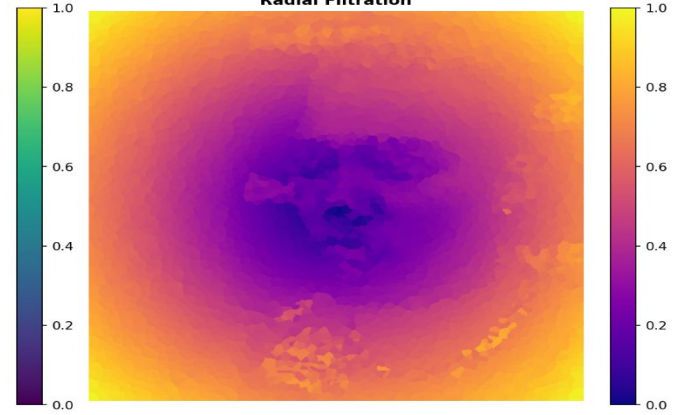
Original Image



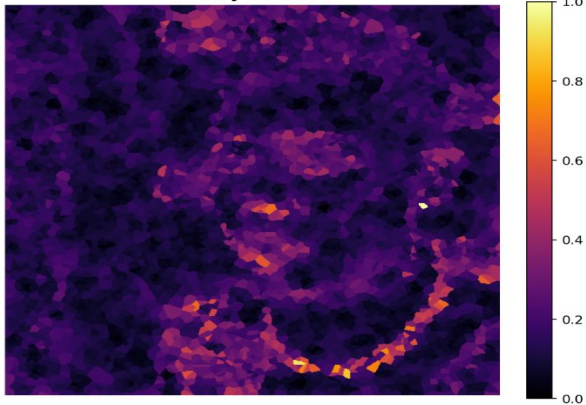
Height Filtration



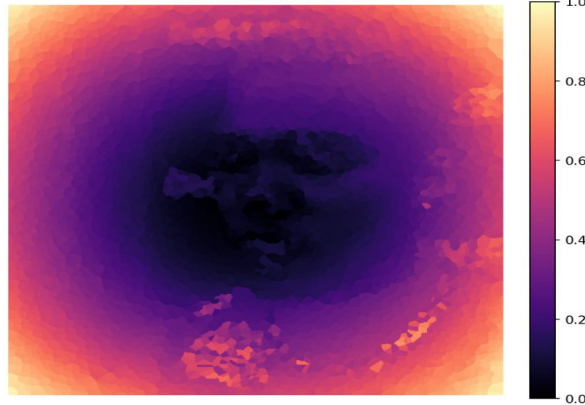
Radial Filtration



Density Filtration



Dilation Filtration



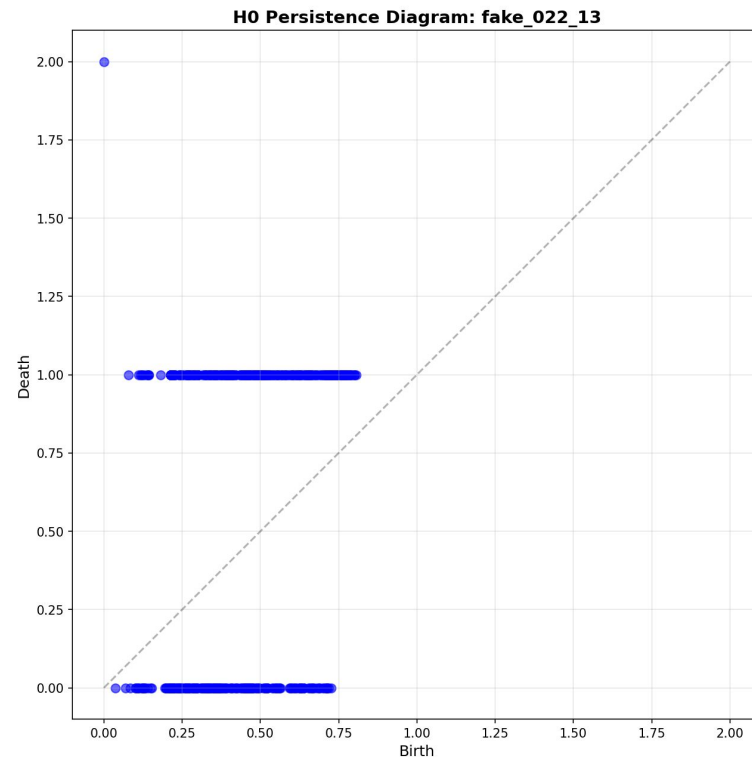
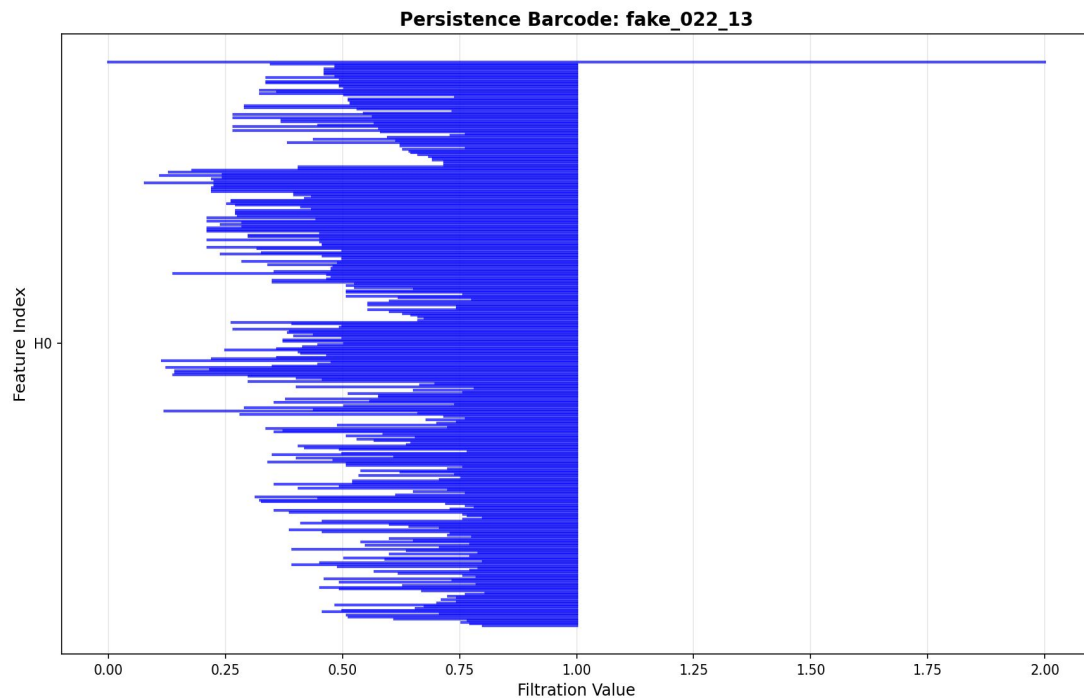
Filtration Analysis
Complete



TDA Feature Extraction

- The primary extracted features are homology groups H_0 (*connected components*) and H_1 (*holes*) *
- By taking the $\beta_0 = \dim(H_0)$ and $\beta_1 = \dim(H_1)$
- Using *giotta-tda*, the birth and death of these homology groups are tracked through filtrations steps in persistence diagrams
- These persistence diagram are then vectorized into persistence landscapes
- This allows for feature extraction into our machine learning model

H0 Diagrams (*Height Filtration*)





Further Extraction

- Persistence entropy was calculated to measure complexity of diagrams:
 - $H_0 = -\sum p_0 \log(p_0)$
 - High persistence entropy = messy, noisy, complex topology
 - Low persistence entropy = clean, dominant structural features
- Wasserstein Distance was calculated between the completed filtration and empty filtration
 - Deepfake images tend to have more irregular, short-lived features
 - Have a tendency to increase the Wasserstein distance



Feature Extraction Summary

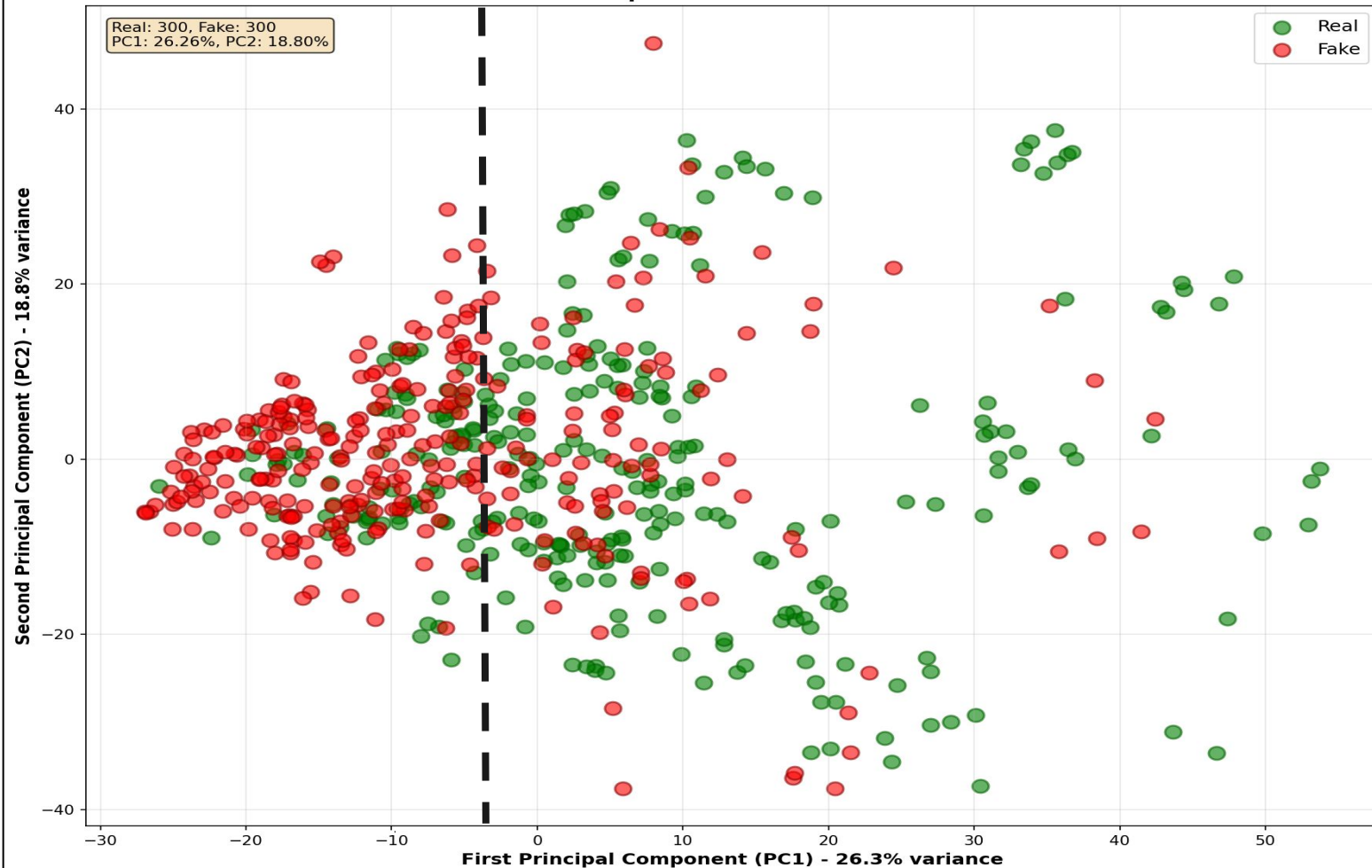
Total Features: 7,224 per image

Feature Sources:

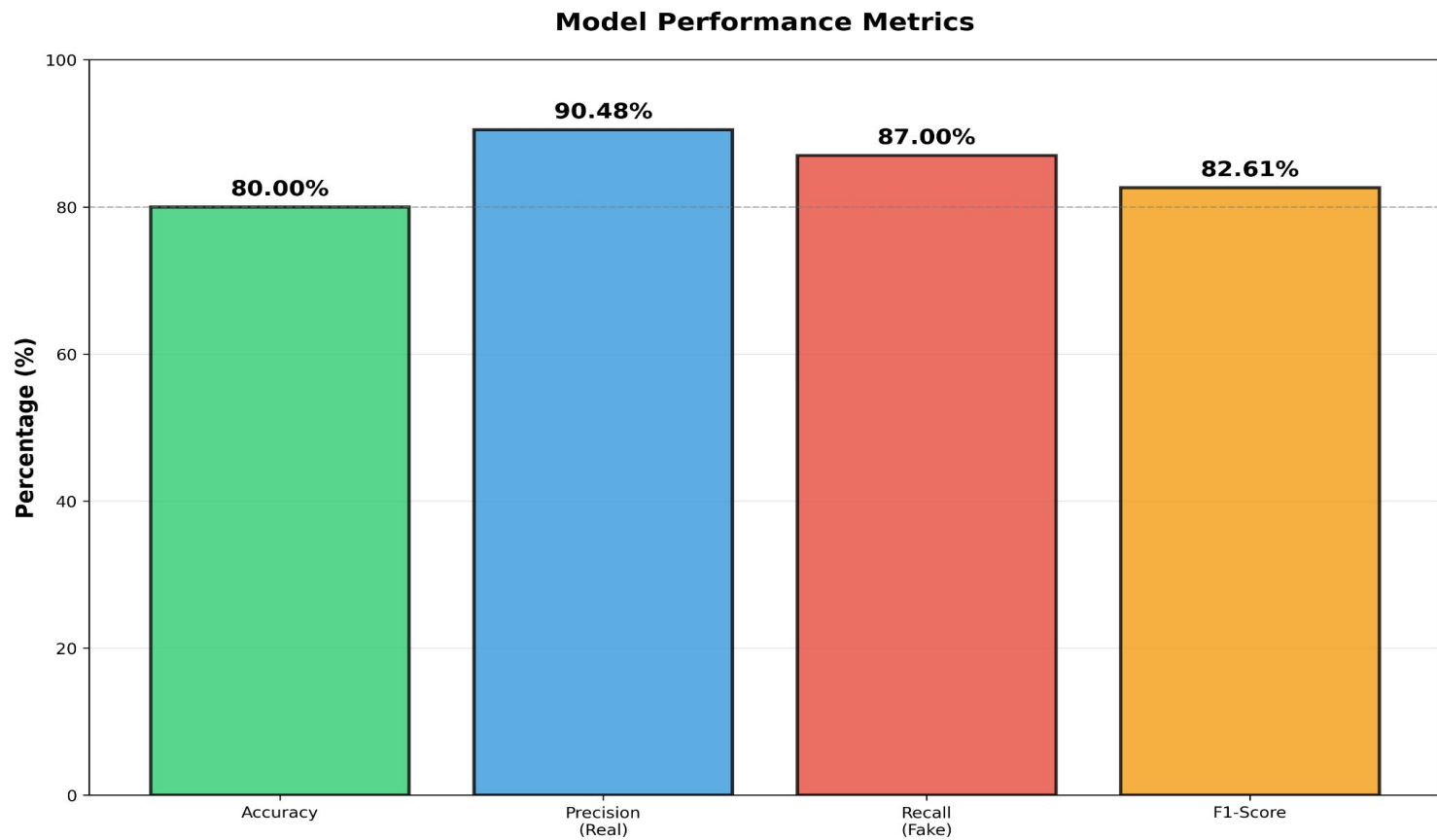
- 4 Filtration Types
 - Height Filtration
 - Radial Filtration
 - Density Filtration
 - Dilation Filtration
- 4 Feature Types per Filtration:
 - Persistence Landscapes
 - Persistence Entropy
 - Wasserstein Amplitudes
 - Betti Curves

2D PCA of Persistence Landscapes - Real vs Fake Images

Total Explained Variance: 45.1%



SVM Results





Limitations

- Computing homology groups is time-intensive; restrict sample size
- Other models may achieve a similar accuracy while being less intensive
- Filtration design choice



Next Steps

- Implementing other filtrations and finding optimal filtrations
- Modifying for deep fake video detection
- Testing it on more noisy and distorted image datasets



Sources

“Jon Ossoff warned about deepfakes. Now he is one.” *The Atlanta Journal-Constitution*, Patricia Murphy, Nov. 12, 2025.

Weaver, N., Logalbo, M., & Pipping, J. (2024). Topological data analysis for classification of AI-generated faces. Undergraduate research project, USPROC/USRESP, University of Florida.

Vaccari, C., & Chadwick, A. (2020). Deepfakes and disinformation: Exploring the impact of synthetic political video on deception, uncertainty, and trust in news. *Social Media + Society*, 6, 205630512090340. <http://dx.doi.org/10.1177/2056305120903408>

https://huggingface.co/datasets/pujanpaudel/deepfake_face_classification